

# Leading a Team in the Age of AI

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Metrics, adoption, and governance for engineering managers

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 **Who is this whitepaper for?**

**Primary audience:** Engineering Managers, Tech Leads, CTOs

**Prerequisites:** Claude Code in active use by the team

**Reading time:** 25 min

**Key takeaways:**

- The 5 current DORA metrics recalibrated for an AI context
- The 5 AI-specific signals that standard frameworks don't cover, plus the agentic metrics DORA misses entirely
- The minimal stack by team size: 5 tools for 5 people, not 20
- The 6 vanity metrics to drop starting tomorrow

**In 5 lines:** AI boosts team velocity, and the numbers are real. The risk is measuring the wrong thing. Sprint velocity, commits, lines of code: these indicators rise automatically when you adopt AI tools, but they tell you nothing about actual code quality or pace sustainability. This whitepaper gives the metrics stack that works when AI generates 70 to 90% of your output, with tools, benchmarks, and an implementation roadmap.

# 1 The Measurement Problem

Adopting AI in development changes delivery speed fast enough to break most existing benchmarks. A team that shipped 2 features per sprint in 2022 can now ship 6, with AI generating 70 to 90% of the code. That looks like a win on every traditional dashboard, and it may genuinely be a win. Or velocity is masking superficial reviews, skill atrophy, and growing technical debt that nobody fully understands.

Activity metrics rise immediately when you adopt AI tools, but quality and maintainability signals are slower and harder to instrument. Sprint velocity, daily commits, lines written: everything goes up. Production bug rate, PR comprehension time, developer confidence: those are the ones that will tell you the truth 6 months from now.

This guide gives engineering managers, tech leads, and CTOs a practical metrics stack: the DORA foundation recalibrated for an AI context, human factors through SPACE, and the AI-specific signals that standard frameworks don't cover.

## 1.1 What Anthropic Data Confirms

Anthropic reported **+67% PRs merged per engineer per day** in January 2026 and 70 to 90% AI-assisted shipped code in an earlier internal study. These figures describe Anthropic's own environment. They are useful context, not an independent benchmark or a target for another team.

Compare one service against its own pre-adoption baseline. If coding time falls but change lead time does not, the constraint is likely in review, validation, or deployment rather than code generation.

## 2 The DORA Foundation

DORA (DevOps Research and Assessment) measures the health of a software delivery system, not individual contributors. The [current DORA guide](#), updated January 5, 2026, defines five metrics grouped into throughput and instability. Apply them to one application or service at a time.

### 2.1 Deployment Frequency

**What it measures:** How often you deploy to production.

**How to measure:** Count production deployments per day, week, or month. Most CI/CD tools expose this directly (GitHub Actions, CircleCI, Vercel, etc.).

**In an AI context:** AI accelerates feature development, so your cadence should increase, provided your pipeline keeps up. If deployment frequency stays flat after widespread AI adoption, the bottleneck is downstream: staging environments, manual QA, review throughput. AI gives you more PRs to merge; it doesn't automatically improve the rest of the pipeline.

**Warning signal:** deployment frequency rising while Change Failure Rate also rises. That's AI-accelerated code that isn't being reviewed seriously.

### 2.2 Change Lead Time

**What it measures:** Time between a commit and that code in production.

**How to measure:** Timestamp at commit, timestamp at deployment. The delta is your lead time. Tools like LinearB and Faros.ai automate this from your CI/CD pipeline.

**In an AI context:** AI reduces coding time but has limited effect on non-code segments of lead time. PR review, staging validation, context-switching delays, and deployment windows are largely unchanged by AI assistance. If your lead time doesn't improve alongside AI adoption, the constraint is in review velocity or pipeline automation, not in coding.

**Diagnosis:** Map your stages explicitly (coding time, waiting for review, waiting for staging, deployment window) to know where the lever is.

## 2.3 Change Failure Rate

**What it measures:** Percentage of deployments that cause a production incident or require a rollback.

**How to measure:**  $(\text{Failed deployments}) / (\text{Total deployments})$ . “Failed” means requiring a hotfix, rollback, or incident response. Define that clearly before measuring.

**In an AI context:** This is the metric most at risk when AI adoption outpaces review discipline. AI generates syntactically correct and structurally plausible code that can still have subtle behavioral errors. Teams that treat AI-generated PRs as “lower risk” and rubber-stamp them see their CFR rise progressively over 6 to 12 months.

**Recommendation:** Track CFR separately for AI-generated code and manually written code. If the AI CFR is significantly higher, your review process needs reinforcement, not your AI tools.

## 2.4 Failed Deployment Recovery Time

**What it measures:** Time to recover from a failed deployment that requires immediate intervention.

**How to measure:** Link the failed deployment, incident, intervention, and recovery timestamp. Exclude failures unrelated to a software change.

**In an AI context:** AI-assisted diagnosis can reduce recovery time when the model can inspect deployment metadata, logs, traces, and code. Measure the recovery-time distribution, false diagnoses, and human intervention time. A demo is not evidence of improvement.

## 2.5 Deployment Rework Rate

**What it measures:** The ratio of deployments that are unplanned work caused by a production incident.

**How to measure:** Divide unplanned corrective deployments by total deployments over the same period, with each corrective deployment linked to its incident.

**In an AI context:** Generation speed can hide hotfixes, rollbacks, and follow-up deployments. Attach token cost and human review time to this rework. A higher deployment frequency with a rising rework rate is not a throughput improvement.

## 2.6 From Four Keys to Five Metrics

The [official history](#) records two changes that many dashboards still miss. DORA replaced MTTR with Failed Deployment Recovery Time in 2023, then added Deployment Rework Rate in 2024. The current model therefore has five metrics.

The 2025 report also replaced the old Elite, High, Medium, and Low clusters with seven broader team archetypes. Historical tier tables are not current universal targets. Compare one service against its own baseline and improve its main constraint.

## 3 Beyond DORA: The SPACE Framework

DORA measures the delivery system; SPACE measures the people inside it. Both are necessary, and neither is sufficient alone.

SPACE was developed by researchers from GitHub, Microsoft, and the University of Victoria (published 2021). It covers five dimensions.

### 3.1 Satisfaction and Well-being

Are developers satisfied with their work, their tools, and their processes? Are there burnout signals?

**Measure with:** a quarterly developer experience survey (5 to 8 questions, anonymous). Track the trend over time, not the absolute score. A team at 3.2/5 moving to 3.8/5 over 6 months is in a better position than a team stuck at 4.0/5.

### 3.2 Performance

Does the work delivered function as intended? Does it meet quality and reliability expectations?

**Measure with:** Change Failure Rate (overlap with DORA), Bug Escape Rate (bugs reaching production divided by total bugs), and customer satisfaction on specific features.

### 3.3 Activity

What volume of work is produced?

**Measure with:** deployment frequency, throughput (features delivered per cycle), PR merge rate.

**Critical note:** Activity is the easiest dimension to measure and the easiest to optimize perversely. High commit counts, high PR volumes, high deployment frequency can coexist with very low actual value delivered. Activity metrics are inputs, not outcomes.

## 3.4 Communication and Collaboration

Is knowledge flowing? Are teams unblocked and connected?

**Measure with:** PR review latency (time between opening and first review), cross-team dependency resolution time, new contributor onboarding time.

## 3.5 Efficiency and Flow

Can developers do deep work without constant interruption? What friction exists in the development process?

**Measure with:** self-reported flow state frequency (in your developer survey), context-switching frequency, unplanned versus planned work ratio.

## 3.6 The Velocity Trap

Teams can reach “High” on DORA deployment frequency while scoring poorly on satisfaction, well-being, and efficiency. More deployments, but developers working evenings to meet sprint commitments, skipping design discussions because AI makes coding fast enough to skip planning, accumulating cognitive debt from AI code reviews they don’t fully understand.

SPACE captures these signals where DORA doesn’t. Using both frameworks together gives you the complete picture, and a divergence (DORA strong, SPACE weak) is an early warning signal of future DORA degradation: exhausted teams eventually deliver more slowly.

### Recommended cadence

**DORA:** monthly leadership review. System health, delivery pipeline performance.

**SPACE** (satisfaction and efficiency dimensions): quarterly review. Human health, sustainable pace, skill development.

## 4 AI-Specific Metrics

Standard frameworks weren't designed with AI-assisted development in mind. These metrics fill the gap.

### 4.1 % AI-Assisted Code

The proportion of committed code that was AI-generated or AI-assisted. Available in Anthropic Contribution Metrics (Team and Enterprise plans), the GitHub Copilot dashboard, and similar tools for other AI coding assistants.

**Why it matters:** Provides context for everything else. A 5% CFR increase is a different signal if AI assists 10% of your code versus 80%. Track this as the denominator for all quality metrics. This number typically rises over time as adoption grows; benchmark it quarterly.

### 4.2 CFR: AI Code vs. Manual Code

Separate your Change Failure Rate by code origin: AI-generated commits versus manually written commits. Most enterprise AI coding tools can tag commits or PRs.

If the AI CFR is within 2 to 3 percentage points of the manual CFR, your review process is working. If it's significantly higher, review discipline has dropped. If it's lower, AI tools may be genuinely improving code quality in your domain.

### 4.3 Review Time: AI PRs vs. Manual PRs

Compare the average review time (opening to merge) for AI-generated PRs versus manually written PRs. If AI PRs are merged significantly faster than manual ones, you may have a rubber-stamping problem.

AI-generated code requires at least as much scrutiny at review as manually written code, possibly more: it can be wrong in non-obvious ways with high apparent confidence. A 30% faster review cycle for AI PRs is a yellow signal worth investigating.

## 4.4 Developer Code Understanding

A binary qualitative signal: during a PR review, can the author explain their AI-generated code in their own words, not just what it does but why it does it that way?

Track this informally through your code review culture. If reviewers start noticing that authors can't explain their AI-generated submissions, that's a skill atrophy signal that can later appear as higher CFR and longer failed deployment recovery time.

## 4.5 PR Comprehension Time

A proxy for code clarity and maintainability: ask reviewers to report how long it took them to understand what a PR does (before they can assess whether it's correct). Track the median across your team.

Rising comprehension time suggests code is becoming more complex or less well-organized over time, regardless of who wrote it. AI-generated code can inflate this metric by producing syntactically dense implementations that are harder to reason about than simpler, more explicit alternatives.

## 5 Agentic Metrics: What DORA Doesn't Measure

DORA and SPACE were designed for deterministic software systems. Agents introduce non-determinism, probabilistic quality, and failure modes that fall through every existing metric category. Three groups of metrics fill that gap.

### 5.1 Why Standard DORA Is Insufficient for Agent Workflows

DORA measures pipeline health, not output correctness. A study of 39 agent frameworks and 439 agentic applications (arXiv 2509.19185) found that 70% of test effort concentrates on deterministic components (tools, workflows) while less than 5% covers the LLM Plan Body, the central reasoning component most likely to produce incorrect results. Most DORA tooling has the same blind spot.

DORA doesn't capture the cost of the verification loop either. Agent-generated PRs require at least as much review scrutiny as manually written code, and in practice more, because subtle behavioral errors can appear in syntactically correct code. Standard deployment frequency and lead time metrics look identical whether review is thorough or rubber-stamped.

### 5.2 Group 1: RCT-Verifiable Metrics

These replicate what METR and DeputyDev measured. Run them on real tasks, not synthetic benchmarks. They give you numbers calibrated to your own team rather than vendor estimates.

| Metric                                           | What it measures                                                     | Published baseline                                                                                                                                                             |
|--------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task completion time, AI vs. no AI               | Actual speedup or slowdown for your task type                        | METR Study 1 (July 2025, arXiv 2507.09089): –19% for experienced developers on complex open-source repos. Perception gap: +39 points, developers believed they were 20% faster |
| PR cycle time, AI-assisted vs. baseline          | Pipeline throughput change                                           | DeputyDev (arXiv 2509.19708): –31.8% PR cycle time (p=0.0018), n=300 engineers, 12 months                                                                                      |
| Pass rate on PRs with executable tests as oracle | How often agent-generated code actually passes functional validation | c-CRAB (arXiv 2603.23448, March 2026): Claude Code 32.1% pass rate on real PRs. Union of four tools: 41.5%. This is the practical ceiling for current AI code review quality   |

The METR –19% figure applies specifically to L1-L2 workflows (Cursor Pro and Claude 3.5/3.7 Sonnet on complex existing repos). It is the only rigorous control-group measurement available for this context. Self-reported figures from McKinsey (20-45%), BCG (64%), and GitHub’s own studies are not replicated under controlled conditions and should be treated as aspirational estimates, not planning inputs.

### 5.3 Group 2: Agentic Pipeline Metrics

These require instrumentation via Langfuse, Arize Phoenix, or AWS Bedrock AgentCore. They do not replace DORA; they sit alongside it as a layer specific to non-deterministic systems.

| Metric                        | What it measures                                                                                                                                                | Reference                                                                                                                                                                                                            |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Spec quality score            | Completeness and precision of specs before agent execution, a leading indicator for output quality                                                              | No standard rubric yet; define internally. Factory.ai's pre-implementation validation contracts are the closest documented proxy                                                                                     |
| Validation contract pass rate | Percentage of agent-generated implementations that pass pre-defined behavioral contracts, measured before human review                                          | Factory.ai Missions pattern: 81 problems detected in a Slack clone from spec alone, generating 34% of implementation work as fix features                                                                            |
| Agent task completion rate    | Tasks the agent completes without human correction, expressed as a percentage                                                                                   | Instrument via harness logs. Anthropic's internal data shows the 99.9th-percentile task duration grew from 25 to 45 minutes between October 2025 and January 2026, indicating agents are handling more complex tasks |
| Code review recall            | Rate at which agent-generated review comments are acted on by developers                                                                                        | Code Review Bench (Martian, March 2026, 200,000+ open-source PRs): Augment Code 62.8% recall, GitHub Copilot 53.3% recall, Graphite 75% precision but only 8.8% recall                                               |
| Cost per completed task       | Token spend plus human review time per agent task that reaches a mergeable state                                                                                | No industry benchmark published yet. Track manually: tokens consumed, cost per model call, human review hours per task completion                                                                                    |
| Tokens per feature            | Average tokens consumed per merged feature, crossed with Jira or Linear ticket boundaries. A better signal than tokens per request because it accounts for ses- | No industry benchmark. Track via cboard's project leaderboard (tokens per session times ses-                                                                                                                         |

## 5.4 Test a Harness Before Standardizing It

A harness trial should compare two or three candidates on the same real tasks, with the same instructions and model where possible. Write pass criteria before the trial, isolate each candidate in its own worktree, and retain the traces that explain the result. A public benchmark can inform the shortlist; it cannot replace that local test.

| Measurement        | Decision question                                                   |
|--------------------|---------------------------------------------------------------------|
| Human verdict      | Is the change acceptable without unplanned re-work?                 |
| Interventions      | How many human corrections or redirects were needed?                |
| Plan drift         | Did the work stay within the stated scope and contract?             |
| Wall time          | What elapsed time passed from launch to verified result?            |
| Accepted-task cost | What cost belongs to tasks the team actually accepted?              |
| Recovery           | What remains after a failure, interruption, or rejection?           |
| Setup friction     | Which permissions, dependencies, and operations remain to maintain? |

The [Agent Harness Map test protocol](#) connects these measures to the choice between a runtime, repository contract, and orchestrator. They complement DORA; they do not measure people.

## 5.5 Group 3: Agent Governance Metrics

Sourced from Strata Identity Research 2026 and CSA/Zenity 2026. These reflect the state of organizational maturity rather than technical performance.

| Metric                             | Target state                                                 | 2026 baseline for context                                                                                                                |
|------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Agents with named human sponsor    | Every active agent has an identifiable owner                 | Only 28% of organizations can link agent actions to a human sponsor in all environments (Strata 2026)                                    |
| Real-time agent inventory coverage | All active agents appear in a central registry               | 21% of organizations maintain a real-time inventory (Strata 2026)                                                                        |
| Credential rotation frequency      | Agent credentials rotate at least every 90 days              | 44% of organizations still use static API keys for agent authentication (Strata 2026)                                                    |
| Permission violations detected     | Violations detected as a share of estimated total violations | 53% of organizations have experienced an agent incident in the past 12 months; 58% took more than 5 hours to detect it (CSA/Zenity 2026) |

## 5.6 The Heavy-User Review Time Contradiction

Digital Applied Q1 2026 (n=2,847 developers) found that heavy AI tool users spend 14-16 hours per week reviewing AI-generated code, compared to 11.4 hours per week for average users. This directly contradicts the narrative that AI reduces review burden. Per-review unit efficiency may improve, but the volume of generated code grows faster than review capacity, the most likely explanation for the gap. Before committing to time-savings projections, measure your team's actual review time distribution across AI-generated and manually written PRs.

Review effectiveness has a ceiling that predates AI and hasn't moved: detection degrades sharply past roughly 200-400 lines per review, or 60-90 minutes of sustained reviewing (Cisco/SmartBear, Cohen 2006). A 14-16 hour weekly review load is therefore not 14-16 hours of equivalent-quality review. See [The Attention Cost of the Review Shift](#) for the underlying studies and the practices that keep review inside the effective band.

## 5.7 The Uplevel Copilot Study

Uplevel's study of Copilot adoption (uplevelteam.com, "genai-developers") found no significant change to coding speed, PR cycle time, or throughput after teams adopted Copilot,

alongside a 41% increase in bug rate within PRs. Their “Sustained Always On” proxy for burnout risk, built from after-hours and weekend activity patterns, also declined more for developers without Copilot than for those with it over the same period.

Read this alongside the heavy-user review time contradiction above: two independent measurements now point the same direction. Neither speed nor throughput moves the way the adoption narrative predicts, and the metrics that do move (bug rate, review time, burnout proxy) move in the wrong direction. Treat any AI-tooling ROI claim resting on developer speed alone as unverified until you’ve checked it against your own bug rate and review-time data.

## 5.8 pass<sup>k</sup> for Non-Deterministic Tests

Standard pass@1 is insufficient for agent-generated code. A test that passes once may fail on the next run because the output is non-deterministic. Promptfoo and LangChain both document the pass<sup>k</sup> pattern: run critical tests  $k$  times consecutively, typically 3 to 5. A test passes only if it passes all  $k$  runs. This is not flaky-test detection, it is a deliberate quality gate for probabilistic systems. Apply it specifically to agent-generated suites, not to the full regression suite where the overhead would be prohibitive.

## 6 Product Metrics

Engineering metrics measure how code is built. Product metrics measure whether the code actually solves the right problems. Most engineering teams track the former and leave the latter entirely to product managers. That creates a blind spot where a team can ship fast, with strong DORA scores, while the product drifts away from user needs.

### 6.1 Time-to-Value

How long does it take a new user to reach their first success with your product? Define “first success” concretely: first task completed, first item saved, first report generated.

Track this as a median across user cohorts, and watch for regressions after major releases. AI can accelerate your feature delivery rate without improving the new user experience, or even degrading it.

### 6.2 Feature Adoption Rate

Among users who could use feature X, what percentage actually uses it within 14 days of release? A feature delivered on time with strong DORA metrics but that nobody uses is still a failed feature.

Segment by user cohort (new versus existing users, different pricing tiers) to distinguish adoption problems from discoverability problems.

### 6.3 Bug Escape Rate

Bugs found in production divided by total bugs (pre-production bugs plus production bugs).

Formula:  $\frac{\text{bugs\_in\_prod}}{(\text{bugs\_before\_prod} + \text{bugs\_in\_prod})}$  .

If your Bug Escape Rate exceeds 20%, your QA and review processes are systematically failing to catch problems before they reach users. With AI-assisted development, this metric deserves close monitoring: faster code generation combined with less rigorous reviews can push Bug Escape Rate up even when the absolute number of bugs stays flat.

## 6.4 Feature CSAT

Customer satisfaction score tied to specific features, not the product as a whole. More actionable than NPS: instead of “how likely are you to recommend us,” ask “how useful was feature X for accomplishing task Y” on a 1-to-5 scale.

NPS is useful for brand-level sentiment but too lagging and too broad to drive development priorities. Feature CSAT gives you a signal within 2 to 4 weeks of a release rather than 6 to 12 months.

## 7 By Team Size

Different team sizes have different tolerances for measurement overhead. A 5-person team spending 20% of its time on metrics infrastructure is making a bad trade-off. A 25-person team without automated DORA tracking is navigating blind.

### 7.1 5-Person Team

**Metrics to track:**

| Metric                                    | How                                           | Frequency                |
|-------------------------------------------|-----------------------------------------------|--------------------------|
| Deployment frequency                      | Manual count or from CI                       | Weekly                   |
| Cycle time (commit to prod)               | Linear or GitHub time-stamp diff              | Per PR, reviewed monthly |
| Time-to-value (north star product metric) | Analytics tool (Mixpanel, Amplitude, PostHog) | Monthly                  |
| Production bugs per month                 | Count in your issue tracker                   | Monthly                  |
| Developer satisfaction                    | Anonymous 5-question form                     | Quarterly                |

**Tooling:** GitHub Insights plus a shared spreadsheet. No need for a dedicated dashboard at this size; the overhead isn't worth it. What matters is the discipline of reviewing these numbers in a monthly retrospective, not the precision of tooling.

#### The 'we're too small' trap

The instinct at this size is often to skip metrics entirely (“we’re too small, we know each other, we talk every day”). Resist that. The value of metrics at 5 people is the discipline, not the visibility. Naming a north star metric and checking it every month forces conversations that daily standups don’t naturally trigger.

## 7.2 25-Person Team

### Metrics to track:

| Metric                           | How                                                       | Frequency          |
|----------------------------------|-----------------------------------------------------------|--------------------|
| All 5 DORA metrics               | LinearB or Faros.ai automated, after checking definitions | Weekly (automated) |
| Cycle time by squad (not global) | Same tooling, segmented                                   | Weekly             |
| Bug Escape Rate                  | Issue tracker plus deployment markers                     | Monthly            |
| Feature CSAT                     | In-app survey on key features                             | Per release        |
| % AI-assisted code               | Anthropic Contribution Metrics                            | Monthly            |
| Developer satisfaction           | Anonymous 8-question survey                               | Quarterly          |
| PR review time                   | GitHub Analytics / LinearB                                | Weekly             |
| Time-to-value                    | Analytics tool                                            | Monthly            |

**Tooling:** LinearB or Faros.ai for DORA automation, GitHub Analytics for AI contribution data, PostHog or Amplitude for product metrics.

#### Avoid global averages

At 25 people, global averages hide squad-level problems. A team with 3 squads where 80% of incidents come from one squad will show a “Medium” global CFR and miss the signal entirely. Track DORA by squad, not just by organization.

PR review time is particularly revealing at this scale. When the review median exceeds 24 hours, it creates context-switching overhead: developers move to other tasks while

waiting, then have to reload context when the review comes back. That reload cost doesn't show up in any standard metric, but it accumulates across dozens of PRs per week.



## 8 Vanity Metrics to Drop

| Drop               | Replace with                             | Why                                                                                                         |
|--------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Sprint velocity    | Cycle time + deployment frequency        | Velocity is manipulable in 2 sprints by changing estimation practices. Cycle time is harder to game.        |
| Lines of code      | Bug Escape Rate                          | LOC measures output volume. With AI, LOC rises automatically. Bug Escape Rate measures output quality.      |
| NPS alone          | CSAT + time-to-value                     | NPS is a lagging brand signal, not an engineering steering metric. Feature CSAT is actionable within weeks. |
| Commits per day    | Lead Time for Changes                    | Commit frequency measures activity. Lead time measures whether that activity actually delivers value.       |
| Story points       | Throughput (features delivered)          | Points are defined relative to the team's own baseline and are manipulable by repointing.                   |
| Code coverage rate | Mutation testing score + Bug Escape Rate | Coverage tells you how many tests exist. Mutation testing tells you how many tests actually deliver value.  |

**Note on story points in an AI context:** if AI generates boilerplate and scaffolding automatically, the effort to implement a “3-point story” has dropped considerably. Teams that haven’t recalibrated their pointing will see velocity increases that reflect tool efficiency, not team capacity.

## 9 The 4-Question Test

Before adding a metric to your stack, run it through these four questions.

### 1. Can you act on it within 2 weeks?

If the answer is no, it's a reporting metric, not a steering metric. Reporting metrics belong in quarterly investor decks, not weekly team reviews. "Total API calls since launch" is a reporting metric. "API error rate over the last 7 days" is a steering metric.

### 2. Does it explain the why, not just the what?

"Churn is at 5%" tells you nothing. "80% of churned users never completed their first workflow" tells you where to look. When evaluating a metric, ask: if this number moves, do I know what to investigate?

### 3. Is it correlated with a business outcome?

Deployment frequency correlates with revenue in fast-iteration SaaS products. PR review time correlates with developer satisfaction and retention. Feature CSAT correlates with expansion revenue. Lines of code correlate with nothing important.

### 4. Can it be measured automatically?

If collecting the metric requires manual work, it will be abandoned within 3 months when load increases. Automate or eliminate.

**The rule:** fewer than 3 "yes" answers, drop the metric. A stack of 5 rigorous metrics is more useful than a stack of 20 poorly defined ones.



# 10 Tooling

| Tool                             | What it does                                                 | Best for                            | Notes                                                                                                                                                   |
|----------------------------------|--------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| LinearB                          | DORA automation + cycle time, connected to GitHub + Jira     | 25+ people                          | Strong DORA dashboards out-of-box, solid PR analytics                                                                                                   |
| Faros.ai                         | DORA + custom engineering dashboards, open-source core       | 25+ people                          | More configurable than LinearB, heavier setup                                                                                                           |
| GitHub Analytics (Anthropic)     | AI contribution metrics (% AI-assisted code, PR attribution) | Any Claude Code team                | Enterprise/Team plan required                                                                                                                           |
| Sleuth                           | Deployment tracking + change failure rate, DORA-focused      | 10+ people                          | Lightweight, CI/CD-focused, no bloat                                                                                                                    |
| Axify                            | Full engineering metrics suite, DORA + flow + team health    | 15+ people                          | Canadian startup, strong SPACE coverage                                                                                                                 |
| GitHub Insights                  | Basic activity metrics, free, native                         | Any size                            | Sufficient for 5-10 people, insufficient at scale                                                                                                       |
| Spreadsheet                      | Manual tracking, always works, zero setup                    | Fewer than 10 people                | The right tool when automated setup overhead isn't justified                                                                                            |
| LiteLLM + Loki + Tempo + Grafana | Self-hosted usage, cost, and trace aggregation               | 25+ people, self-hosted requirement | LiteLLM Gateway logs usage server-side with zero client-side setup; Loki ingests session JSONL for file-level detail; Tempo correlates AI sessions with |

No tool automatically surfaces the AI-specific metrics described above (CFR by code origin, review time comparison, comprehension signals). That requires either custom dashboards built on your CI/CD data or manual tracking. GitHub Analytics covers % AI-assisted code; the rest you'll need to wire up yourself. Teams that can't send usage data to a third-party vendor should default to the self-hosted LiteLLM/Loki/Tempo/Grafana path instead of LinearB or Faros.ai.

**Avoid tool proliferation.** A team with LinearB, Jira, GitHub Insights, and two separate analytics tools will spend more time reconciling numbers than acting on them. One DORA tool, one product analytics tool, GitHub Analytics for AI-specific data.

## 10.1 Broader Delivery Intelligence Platforms

The starter table above covers what most teams reach for first. The 2026 market is wider, and worth a second look if that table doesn't fit your organization's size or constraints.

| Tool                       | What It's For                                                                                                                                   | Notes                                                                                                                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DX (getdx.com)             | Developer intelligence platform built by researchers associated with the DX Core 4 / SPACE framework lineage                                    | Positioned for larger organizations wanting a research-grounded metrics platform rather than a plug-and-play dashboard                                                                              |
| Multitudes (multitudes.ai) | Analytics and recommendations built directly on DORA, SPACE, and DevEx signals                                                                  | Sends “nudges” when a metric indicates a team is blocked or overloaded, rather than just reporting the number                                                                                       |
| Swarmia (swarmia.com)      | Cycle time breakdown (shipped 2022, still active): decomposes PR time into in-progress, waiting-for-review, and waiting-for-merge               | Also publishes its own research on the productivity impact of AI coding tools (2025 blog)                                                                                                           |
| Cortex.io                  | Positions itself as an “Engineering Operations Platform” and publishes its own “Engineering Intelligence Platforms: Top 8 Tools” category guide | Already cited elsewhere in this whitepaper series as a data source for PR size and change failure rate figures; documented here as a delivery-intelligence tool in its own right for the first time |
| Jellyfish                  | Positions itself in 2026 as a “software engineering intelligence platform” explicitly built for AI-integrated organizations                     | Same note as Cortex.io: already cited as a data source elsewhere, now documented as a tool                                                                                                          |
| Obeya (obeya.io)           | Aggregates 20+ existing DevOps tools into a single layer, rather than being a standalone data source itself                                     | Fits teams documenting specific DevOps tool state                                                                                                                                                   |

## 10.2 AI-Generated Board Narratives

In 2026, some of these platforms now generate a written narrative from the metrics they already compute, rather than leaving that translation to a human, a genuinely new capability. Two concrete examples.

**LinearB** generates an AI iteration summary from Jira, Git, PR, and deployment data, structured around three axes: what went well, what didn't, and what could improve. It's delivered automatically to Slack or Microsoft Teams at sprint end, replacing the manual retro write-up a tech lead would otherwise produce.

**Jellyfish's "AI Executive Report"** translates AI-tool adoption data (Copilot, Windsurf, and similar) into three dimensions built for a non-technical executive audience: adoption, output, and impact, the last framed as time saved, money saved, and risk reduced.

### What the AI is actually doing

In both cases, the generative AI's role is explanation and prioritization of what the underlying analytics already computed, not the computation itself. The pass rate, the cycle time, the CFR split: those numbers come from the same instrumentation covered throughout this whitepaper. The LLM's job is turning a table into three paragraphs a stakeholder will actually read, not producing new statistical insight.

# 11 Probabilistic Delivery Forecasting

Every benchmark so far in this whitepaper answers “how are we doing.” This section answers a different question: “when will this ship,” and it answers it with a probability distribution instead of a single date.

## 11.1 Why a Point Estimate Misleads

Traditional forecasting takes your velocity (story points or throughput per sprint) and divides it into remaining scope to produce a single date. That date carries false precision, implying a certainty the underlying data never had. Two teams with identical average velocity can have wildly different variance, and variance is exactly what a point estimate throws away.

Monte Carlo forecasting instead runs your team’s actual historical cycle times (or throughput) through thousands of simulated iterations and produces a distribution: “80% confidence between March 3 and March 20” instead of “March 12.” That’s a more honest answer, because it’s the one your data can actually support.

## 11.2 Tools

**ActionableAgile** ([actionableagile.com](https://actionableagile.com)) is a flow and predictability-focused analytics product, licensed via 55 Degrees AB, built around Monte Carlo forecasting. Its “Portfolio Forecaster” extends the same technique across multiple initiatives at once, useful when a board is asking about predictability across a roadmap rather than a single deliverable.

**Nave** ([getnave.com](https://getnave.com)) runs Monte Carlo simulation alongside a “Cycle Time Histogram” view.

### The one requirement that makes or breaks this

Nave’s own documentation states the core constraint plainly: “The sole requirement for Monte Carlo to work and give reliable answers is to use data produced by a predictable delivery system.” Even 20-30 completed items are enough if the workflow is stable, but a recent change to team composition or process should reset the data window so the forecast reflects current conditions, not a mix of old and new reality.

## 11.3 What This Does Not Fix

Monte Carlo forecasting is not a fix for an unstable or unpredictable delivery system. If your cycle times swing wildly because scope keeps changing mid-sprint, or because the team composition just shifted, the simulation will faithfully replicate that unpredictability as a wider distribution. It won't narrow the range for you. A wide, honest 80% confidence interval is still more useful than a false point estimate, but it's a symptom report, not a cure. Fix the underlying instability first; the forecast will narrow on its own once the process does.

# 12 Implementation Roadmap

The most common failure mode in metrics programs is trying to instrument everything at once.

## 12.1 Phase 1 (Weeks 1-2): Instrument DORA

Connect your CI/CD pipeline to a metrics tool. For most teams, that means connecting GitHub Actions or its equivalent to LinearB, Faros, or Sleuth. Automate Deployment Frequency and Change Lead Time first. Change Fail Rate, Failed Deployment Recovery Time, and Deployment Rework Rate require deployment-to-incident linkage.

**Output:** a live dashboard showing at minimum Deployment Frequency and Lead Time for Changes. Your first baseline numbers.

## 12.2 Phase 2 (Weeks 3-4): Baseline and Targets

Once you have 2 to 4 weeks of data, establish your real baseline. The temptation here is to compare immediately to industry benchmarks. Resist that. Set internal improvement targets first: “reduce Lead Time by 20% over the next quarter” is more actionable than “reach the High tier.” Your context (tech stack, deployment environment, team size, product type) affects what’s achievable more than any industry benchmark.

Launch your first developer satisfaction pulse (5 questions, anonymous, 10 minutes to build in Google Forms or Typeform). That’s your SPACE baseline.

## 12.3 Phase 3 (Month 2+): Product and AI Metrics

Once DORA is stable and automated, add product metrics (time-to-value, Feature CSAT) and AI-specific signals (% AI-assisted code, CFR by origin). These require more setup (product analytics instrumentation, PR tagging conventions) but are worth the investment once your DORA foundation is solid.

Review the full stack quarterly and prune ruthlessly. Any metric that hasn’t guided a decision in the last 3 months is a reporting metric disguised as a steering metric. Cut it.

## 13 Reporting Delivery Capacity to a Skeptical Board

Everything above this section assumes the audience is your own team or your engineering leadership chain. A board that has watched past estimates slip needs a different approach, because the problem it's raising usually isn't the one a dashboard answers.

### 13.1 The Diagnostic: Doubt Is a Trust Problem, Not a Data Problem

When a board starts doubting a team's delivery capacity after past estimates have slipped, adding more delivery dashboards rarely fixes it on its own. The board's doubt is almost always about trust and visibility, about what caused the slip and whether it will happen again. It is much less often about the data itself: board members generally don't disbelieve the velocity number on the screen, they disbelieve the story that number is supposed to support.

#### An evidence gap, stated plainly

No published study or vendor report actually measures whether delivery-intelligence tooling repairs executive trust. That's a gap in the field, not a gap in this whitepaper. Nothing here, and none of the tools covered earlier, comes with proof that it fixes a board's confidence. What follows reframes the conversation into one a board can act on, not a guaranteed remedy.

### 13.2 Bring Scenarios, Not a Velocity Chart

A single velocity number invites a single question, whether the team can go faster. That's rarely the question worth answering, and it's an easy one to lose regardless of the real numbers behind it.

Bring 2 to 3 named delivery scenarios instead, each with a concrete timeline, a rework or cost estimate if sequencing gets compressed, and the business impact of picking that option. "Ship the full scope in Q3 at current pace" versus "compress two workstreams into Q2 at an estimated higher defect rate and additional engineer-weeks of rework" versus "cut scope and ship on the original date" is a decision a board can actually make. "Can the team go faster" is not.

## 13.3 Cap Strategic Objectives, Don't Roadmap Every Feature

A fully detailed, multi-quarter feature roadmap looks like rigor and produces the opposite. Every feature-level commitment beyond roughly one quarter out is a precision claim your delivery system almost certainly can't back, and it erodes trust the moment reality diverges from the plan, which it will.

Cap the number of board-level strategic objectives instead: a small, named set, reviewed quarterly. Objectives survive a schedule slip better than feature line items do, because an objective describes an outcome the team is still steering toward even when the specific path changes.

## 13.4 Separate Committed Scope from Best-Effort Scope, and Track the Hit Rate

Draw an explicit line between what the team commits to and what it will attempt on a best-effort basis. Then track, over time, how often committed scope actually ships as committed.

That hit-rate track record becomes the trust-rebuilding metric in its own right. A board that sees "when this team commits, it delivers" repeated release after release cares about that pattern more than it cares about any single throughput or velocity number. Trust rebuilds through a track record, not through a chart.

## 13.5 Align Stakeholders Before the Room, Not in It

A plenary board meeting is a poor venue for first-time alignment on a contentious delivery question. Before walking in with scenarios and numbers, talk to board members individually and understand what's actually driving each one's doubt: comparison to other teams in the portfolio, external business pressure unrelated to engineering, or a specific past miss they haven't let go of. Those three causes call for different framing, and a plenary meeting doesn't give you the space to diagnose which one you're dealing with in the room.

## 13.6 Where Forecasting and AI Narratives Fit, and Where They Don't

The probabilistic forecasting tools covered earlier in this whitepaper give you the honest range behind each scenario above: “80% confidence between these two dates” is a more defensible number to bring into the room than a single point estimate, provided your delivery system is stable enough for the forecast to mean something. The AI-generated board narratives from tools like LinearB and Jellyfish reduce the manual effort of translating sprint data into a written summary a non-technical board member can read.

Neither substitutes for the harder work above. A better forecast and a better-written summary still won't cap your roadmap, separate committed from best-effort scope, or have the individual conversations that surface what a specific board member actually doubts. Those three remain a management job, not a tooling job.

# 14 Summary: The Complete Stack

## Minimal stack by maturity phase

### Phase 1: Foundation (months 1-2)

- Deployment frequency
- Lead Time for Changes
- Developer satisfaction (baseline)

### Phase 2: Quality signal (months 2-4)

- Change Failure Rate (global + by AI/manual origin)
- Failed Deployment Recovery Time
- Deployment Rework Rate
- Bug Escape Rate
- % AI-assisted code

### Phase 3: Full steering (months 4+)

- Time-to-value
- Feature CSAT
- PR review time (AI vs. manual)
- PR comprehension time
- Developer satisfaction (recurring)

The temptation is to want everything from the start. A metrics stack, like a codebase, accumulates debt when it grows faster than the team's capacity to act on it. Build it in phases, prune regularly, and keep the 4-question test as a permanent filter.

## 14.1 PR Audit Trail

For regulated environments and compliance-conscious orgs, a PR audit trail captures four things at PR creation: the tool call log (which tools Claude used and on which files), the list of files modified with before/after line counts, session metadata (session ID, timestamp, Claude Code version, model), and a hash of the active CLAUDE.md rules. A PreToolUse hook writes this to a daily JSONL log, a GitHub Actions step uploads it as a 90-day artifact on each PR, and a periodic compliance script aggregates tool usage and unique files touched across a date range. This gives auditors a structured “what did Claude do during this change” answer without relying on session memory.

## 15 See Also

- [Team Metrics: Reference Guide](#): full version with dashboards and additional resources
- [Session Observability](#): Claude Code session monitoring, cost tracking, usage patterns
- [AI Traceability](#): AI contribution auditing, attribution, and compliance
- [Choosing Your Adoption Approach](#): Claude Code team adoption patterns, CLAUDE.md configuration strategies

# 16 The Series

This whitepaper is part of the **Claude Code Ultimate Guide** series, a comprehensive resource for mastering Claude Code from installation to production deployment.

## 12 whitepapers in the series:

- **WP00** Introduction & Fundamentals
- **WP01** Effective Prompts
- **WP02** Customization & Configuration
- **WP03** Production Security
- **WP04** Internal Architecture
- **WP05** Team Adoption
- **WP06** Privacy & Compliance
- **WP07** Condensed Reference Guide
- **WP08** Agent Teams & Sub-Agents
- **WP09** Learning with AI (UVAL)
- **WP10** AI Budget & ROI
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